

AI511 – Linear and integer programming

Sheet 0, Autumn 2026

Solution:

Included.

Review of elements from Linear Algebra that are used in AI511 - MM505.

Exercise 1

Consider the matrices:

$$A = \begin{bmatrix} 2 & 0 \\ -4 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -7 & 2 \\ 5 & 3 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 9 \\ -3 & 0 \\ 2 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -2 & 1 & 8 \\ 3 & 0 & 2 \\ 4 & -6 & 3 \end{bmatrix} \quad E = \begin{bmatrix} 0 & 3 & 0 \\ -5 & 1 & 1 \\ 7 & 6 & 2 \end{bmatrix}$$

In each part compute the given expression. Where the computation is not possible explain why.

1. $D + E$
2. $D - E$
3. $5A$
4. $2B - C$
5. $2(D + 5E)$
6. $(C^T B)A^T$
7. $2\text{tr}(AB)$
8. $\det(E)$

Exercise 2

Consider the following system of linear equations in the variables $x, y, z \in \mathbb{R}$.

$$\begin{aligned} -2y + 3z &= 3 \\ 3x + 6y - 3z &= -2 \\ -3x - 8y + 6z &= 5 \end{aligned}$$

1. Write the augmented matrix of this system.
2. Reduce this matrix to row echelon form by performing a sequence of elementary row operations.
3. Solve the system and write its general solution in parametric form.

Solution:

```
import numpy as np

# The augmented matrix
AA = np.array([[ 0, -2, 3, 3],
               [ 3, 6, -3, -2],
               [-3, -8, 6, 5]])

In [30]: import sympy as sy
...: # np.linalg.solve(A,b)
...:
...: sy.Matrix(AA).rref()
...:
Out[30]:
(Matrix([
[1, 0, 2, 7/3],
[0, 1, -3/2, -3/2],
[0, 0, 0, 0]]), (0, 1))
```

Hence, the solution is:

$$\mathbf{x} = \begin{bmatrix} 7/3 - 2t \\ -3/2 + 3/2t \\ t \end{bmatrix} = \begin{bmatrix} 7/3 \\ -3/2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 3/2 \\ 1 \end{bmatrix} t \quad t \in \mathbb{R}$$

Exercise 3

Consider the following matrix

$$M = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix}.$$

1. Find M^{-1} by performing row operations on the matrix $[M \mid I]$.
2. Is it possible to express M as a product of elementary matrices? Explain why or why not.

Solution:

```
import numpy as np

M = np.array([[ 1, 0, 1], [-1, 1, 0], [2, 2, 2]])
MM = np.concatenate([M, np.identity(3)], axis=1)

import sympy as sy
sy.Matrix(MM).rref()

(Matrix([
[1, 0, 0, -1.0, -1.0, 0.5],
[0, 1, 0, -1.0, 0, 0.5],
[0, 0, 1, 2.0, 1.0, -0.5]]), (0, 1, 2))
```

Yes, it is possible. Since the matrix is invertible we have shown above that we can go from an identity matrix to M^{-1} and consequently also to M from an identity matrix with elementary row operations. Elementary row operations can be expressed as products between elementary matrices.

Exercise 4

1. Given the point $[3, 2]$ and the vector $[-1, 0]$ find the vector and parametric (Cartesian) equation of the line containing the point and parallel to the vector.

2. Find the vector and parametric (Cartesian) equations of the plane in $\mathbb{R}^3$ that passes through the origin and is orthogonal to $\mathbf{v} = [3, -1, -6]$.

Solution:

The vector equation: Let $[3, 2]^T = \mathbf{p}$ and $[-1, 0]^T = \mathbf{v}$. Any point $\mathbf{x}$ on the line can be expressed as:

$$\mathbf{x} = \mathbf{p} + t\mathbf{v} \quad \forall t \in \mathbb{R}$$

We can derive the Cartesian equation by eliminating t from the equation above:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

From the first coordinate: $x_1 = 3 - t$ and from the second: $x_2 = 2$. The Cartesian equation is $x_2 = 2$ since x_1 is free to get any value.

The plane through the origin orthogonal to $\mathbf{v} = [3, -1, -6]$ is given by:

$$\mathbf{x}^T \mathbf{v} = 0$$

That is: $3x_1 - x_2 - 6x_3 = 0$.

Exercise 5

Write a one line description of the methods you know to compute the inverse of a square matrix.

Solution:

- using cofactors for the adjoint matrix and dividing by the determinant
- by row reduction of $[A|I]$

Exercise 6

Calculate by hand the inverse of

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

and check your result with the function `numpy.linalg.inv`.

Solution:

$$\begin{bmatrix} 0. & 1. & -1.5 \\ 1. & -3. & 4. \\ 0. & 0. & 0.5 \end{bmatrix}$$

Exercise 7

Use Cramer's rule to express the solution of the system $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 3 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

Solution:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is the matrix obtained from A by replacing the i th column with the vector $\mathbf{b}$.

$$x_1 = -4.5 \quad x_2 = 14 \quad x_3 = 1.5$$

Exercise 8

Given two points in the Cartesian plane $\mathbb{R}^2$, $A = (1, 2)$ and $B = (3, 4)$ write the vector parametric equation and the Cartesian equation of the line that passes through them. Express the vector equation as an affine combination of the two points.

Solution:

The vector equation is an affine combination of the two points:

$$\mathbf{x} = [1, 2]^T + t([3, 4]^T - [1, 2]^T), \forall t \in \mathbb{R}^2$$

To find a, b, c such that $ax + by + c = 0$ describes the line we can rewrite the equation above as

$$[x, y]^T = [1, 2]^T + t([1, 2]^T), \forall t \in \mathbb{R}^2$$

we then eliminate t by substituting in the two equations.

Exercise 9

Express the segment in $\mathbb{R}^2$ between the points $A = (1, 2)$ and $B = (3, 4)$ as a convex combination of its extremes.

Solution:

$$\{\alpha[1, 2]^T + \beta[3, 4]^T \mid \alpha, \beta \in \mathbb{R}, \alpha, \beta \geq 0, \alpha + \beta = 1\}$$

Exercise 10

Write a generic vector parametric equation and a generic Cartesian equation of a plane in $\mathbb{R}^3$.

Solution:

$$\begin{aligned} \mathbf{x} &= \mathbf{p} + s\mathbf{v} + t\mathbf{w}, \quad x, t \in \mathbb{R} \\ ax + by + cz + d &= 0 \end{aligned}$$

Exercise 11

Write a generic Cartesian equation of an hyperplane in $\mathbb{R}^n$ that does not pass through the origin.

Solution:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

Exercise 12

Prove that the following vectors in $\mathbb{R}^3$ linearly independent?

- $[6, 9, 5]^T$
- $[5, 5, 7]^T$
- $[2, 0, 7]^T$

Solution:

We need to solve homogeneous system $A\mathbf{x} = \mathbf{0}$, where the matrix A has the three vectors forming its columns. The matrix A has $\det(A) \neq 0$ and rank 3. Therefore the only solution to the homogeneous system is the trivial solution $\mathbf{0}$. The three column vectors are therefore linearly independent.

```
A=np.array([[6, 9, 5],[5, 5, 7],[2, 0, 7]]).T
np.linalg.det(A)
np.linalg.matrix_rank(A)
```